

Temple Inscriptions Recognition and Transliteration in Devanagari Script

Dr. B. Sathish Babu
Department of Artificial
Intelligence and ML
RV College of Engineering
Bengaluru, India
bsbabu@rvce.edu.in

Sannidhi Shetty
Department of Computer Science
and Engineering
RV College of Engineering
Bengaluru, India
sannidhishetty888@gmail.com

Anushka Agarwal
Department of Computer Science
and Engineering
RV College of Engineering
Bengaluru, India
anushka24agarwal@gmail.com

Sai Lahari Sreerama
Department of Computer Science
and Engineering
RV College of Engineering
Bengaluru, India
sailahari2002@gmail.com

Vaishnavi K Bhustali
Department of Computer Science
and Engineering
RV College of Engineering
Bengaluru, India
vaishnavikb296@gmail.com

Sanka Sanjana
Department of Computer Science
and Engineering
RV College of Engineering
Bengaluru, India
sanjanasanka24@gmail.com

Abstract—Ancient inscriptions, palm scripts, manuscripts, etc., have vital information about India's rich culture. Recognition and understanding of these inscriptions have been challenging for epigraphers and professionals. The goal of the proposed research is to advance optical character recognition methods for archival Vatteluttu script inscriptions, which date back to the 4th or 5th century AD. This paper discusses a deep learning model to transliterate the ancient Tamil inscriptions (Vatteluttu Script), which can be extended further to other languages. The proposed work is beneficial to epigraphists, archaeological researchers, and the general public who are interested in this topic. The developed deep learning model has achieved an accuracy of 84.12%.

Keywords—Temple Inscriptions; Ancient Tamil script; Optical character recognition (OCR); Deep Neural networks; Data augmentation; TrueType Font.

I. INTRODUCTION

Access to inscriptions and manuscripts is necessary for gaining a comprehensive understanding of ancient civilizations' culture and history. Ancient temples in India have inscriptions written on rocks, slabs, pillars, temple walls, etc. The majority of these inscriptions provide valuable and extensive information about the monarchy's ancient administrative and religious practices. These inscriptions are regarded as excellently written documentation that helps us to comprehend the standard of living in ancient times. Digitalizing these inscriptions is therefore essential [1].

A few challenges faced by the researchers and epigraphers in deciphering the ancient temple inscriptions are differentiating the foreground pixel from the background gravestone images, rephrasing, perspective deformation, different light illumination, the same kind of background/ focus, damaged monuments, lack of shape and size of the text.

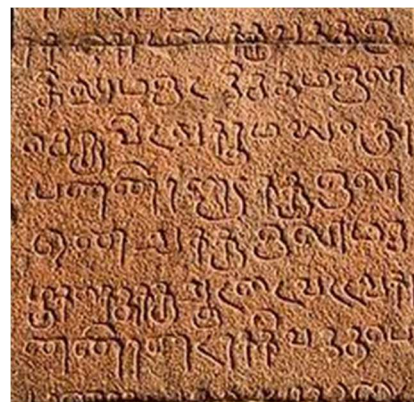


Fig 1.1: Inscriptions (Image Source: [16])



Fig 1.2: Inscriptions (Image Source: [17])

Optical character recognition (OCR) is used to recognize and extract characters from printed or handwritten documents or images [2]. It can aid in preserving inscriptions by generating digital copies that are less prone to physical damage. The digital format allows for easy searchability, comparison, and analysis of the inscription's content. Moreover, OCR can assist in translating and transliterating inscriptions into contemporary languages, making them more accessible to a broader audience. The technology also enhances the overall visitor experience and promotes cultural exchange.

In this research, we have introduced an Optical Character Recognition system that accepts input from any Vatteluttu script inscription, pre-processes it to obtain better visual pictures, and then conducts character-wise segmentation (OpenCV). The transliterated text of the inscription is then produced using the characters fed into a convolutional neural network (CNN) model [3].

II. RELATED WORKS

According to a method suggested by T Manigandan et al. in [4], each letter's image features, such as the count of curves, lines, loops, and dots, are extracted from pre-processed and segmented images of different Tamil characters. Characters are categorized using the Support Vector Machine (SVM) algorithm. The appropriate Unicode value will then be given to each identified character.

Merline Magrina M et al. in [5] proposed a system that includes OCR application modules, i.e., Ensemble Learning. The collected inscriptions are pre-processed and segmented. The output is used in post-processing processes to get rid of additional noises like opening and shutting noises, erosion noises, and dilation noises. Orientation points and magnitude points are recognized using the SIFT technique, and the segmentation procedure is carried out using a bounding box. Characters are categorized using the Ensemble learning classifier, and Unicode is used to match the categorized characters.

Vijayalakshmi. R et al. in [6] describe the various strategies and methods for retrieving, categorizing, and obtaining data from ancient inscriptions written in various languages. The classification is carried out using a variety of classifiers, including SVM, ANN, CNN, and numerous other hybrid algorithms.

To identify the handwritten Tamil characters, Nagul Ulaganathan et al. introduced a novel convolutional neural

network layout [7]. The convolutional neural network (CNN) is used to extract features from handwritten characters. After being trained with a substantial dataset containing thousands of handwritten Tamil characters, the proposed model produced the best results during testing and training.

Lalitha Giridhar et al. in [8] proposed a method for obtaining a sample inscription's digitized output. The methods aimed at pre-processing the raw input image using techniques like binarization, gray scaling, etc., and segmenting them character-wise before feeding into the CNN model. Developing a text-to-speech application and a digitized equivalent current Tamil translation using image recognition and machine learning approaches.

Ruwanmini et al. in [9] proposed the creation of an inscription database and developed a web-based application to assist with character recognition from scanned photographs of inscriptions. The study primarily focuses on employing an appropriate character recognition algorithm to recognize each isolated nature set in these inscriptions. They have discussed Optical Character Recognition processes and technologies.

S. Dhivya et al. in [10] analyzed the various theoretical perspectives adopted by recent studies in the field of identifying ancient Tamil characters from stone inscriptions. An elaborate process that includes pre-processing, feature extraction and classification is provided. Pixel brightness modifications, geometrical transformations, image filtering, and segmentation are a few pre-processing methods. Principal component analysis, linear discriminant analysis, autoencoders, and other feature extraction techniques are used to reduce dimensionality. Convolutional neural networks are used for classification, which comes after all other analysis stages.

Sonika Rani Narang et al. [11] proposed a line segmentation algorithm for the ancient scripts. OCR plays a significant role in digitizing ancient inscriptions and manuscripts. The scanned script is divided into fixed-size vertical stripes. The horizontal Projection Profile (HPP) method is applied to each bar. Every row and each bar's total number of dark pixels is determined. A row is considered a separator if it has fewer than three dark pixels. Consecutive separators are merged. The average line height is used to detect false and missing separators, preventing over-segmentation & under-segmentation.

Munish Kumar et al. in [12] developed a deep learning model for Devanagari ancient character recognition. Their system procedure includes data collection and processing, and normalization. They chose CNN as the classifier and feature extractor and used a dataset consisting of 33 classes of Devanagari script for their experimental work. The authors used 3 CNN layers, namely activation, pooling, and fully connected layer, and achieved an accuracy of 93.73% after 30 epochs. They have also witnessed incorrect recognition due to similarities in shape and faded characters.

III. DATASET

A. Dataset

Dataset played a crucial role in the successful execution of the project. A synthetic dataset was prepared. Images of manuscripts and inscriptions were taken and pre-processed. Then the pre-processed images were segmented and grouped into characters of the Vatteluttu script. This synthetic dataset was then used for training the CNN model. A total of 215 classes were created, and unique codes were given to each class, which helped classify characters.

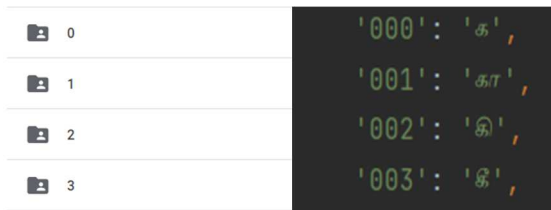


Fig. 3.1: Classification of Vatteluttu characters into classes and its mapping to modern Tamil characters

B. Data Augmentation

The classification model works better with a huge dataset. By making minor adjustments to the existing character images, the data augmentation method is used to expand the training dataset. The methods used are Padding, Gaussian Noise, and a combination of shearing, rotation, and brightness[13].



Fig 3.2: (a) is the original image, (b) and (c) are augmented images

IV. PROPOSED WORK

The proposed method of OCR for Vatteluttu Script can be broken down into four key stages, as shown in Figure 4.1. The first stage is pre-processing the input images. Second,

segmentation of the pre-processed script with the next step, character recognition (feature extraction), and the final step, digitizing using TrueType Font.

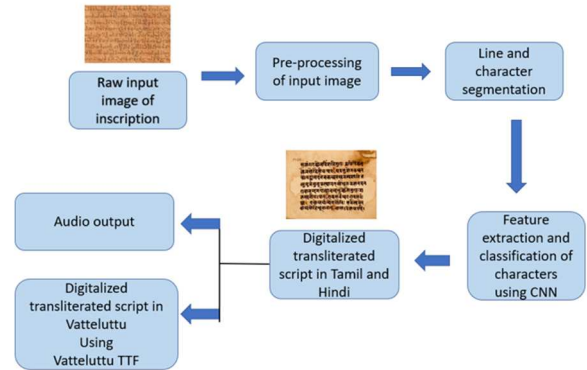


Fig 4.1: Methodology used for transliteration of script

1. Pre-processing

In the proposed approach pre-processing is done to refine the image for better accuracy in later stages. Grayscale conversion of the input image comes first. The picture is then sharpened using a kernel convolution matrix that has the values $\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$ fed into it.

Pre-processing is done further depending on the type of image input given, which is either stone inscription or manuscript. Binarization is a common and crucial technique in both cases. In the suggested work, the Otsu thresholding technique is utilized to binarize the picture if the given input image is a stone inscription. Additional noise reduction also occurs using the "fast non-local means denoising algorithm" with a denoising strength set to 20. In the case of manuscript images, the thresholding algorithm used is binary thresholding, and the output is shown in Figure 4.2. Further Dilation, Erosion operations can be done to improve the binarized image quality.



Fig 4.2: Output image obtained after thresholding image

2. Segmentation

In the suggested study, segmentation [14] comes next to obtain separated image pixels based on semantic content for further display. Segmentation entails dividing the image into horizontal blocks (lines) and vertical blocks (letters). Projection values are first computed using NumPy's sum function to slice the image horizontally. Now, the image's single characters are obtained using a vertical projection

technique, as shown in Figure 4.4 from the image in Figure 4.3. Based on the geometry of the image and the specified percentage, the function determines how much of the border should be removed, removes it from the picture, and then transposes the horizontally divided image. Further, it identifies key rows where a new object is likely to start based on the presence of non-background pixels, which is done by iterating over the rows and checking the difference between adjacent rows. If the difference is more significant than a threshold value (calculated based on the percentage of the image height), then the current row is considered a key row. The final output obtained is each segmented character from the word as shown in Figure 4.5.

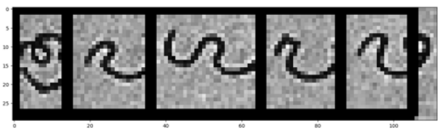


Fig 4.3: Segmented Line

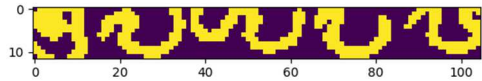


Fig 4.4: Letter-wise segmentation of the input word

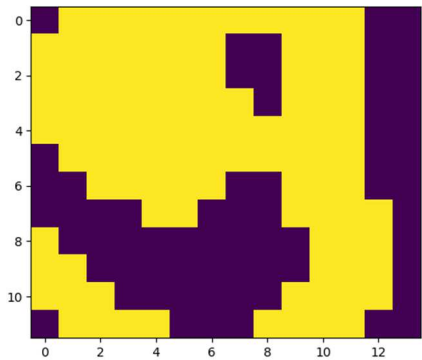


Fig 4.5: Segmented Letter

3. Feature extraction

Features extracted from inscription images by CNN include edges, lines, and characters' shapes, including curves and angles. CNN also identifies the texture of characters, such as the variations in depth and colour. CNN also specifies characters' size and aspect ratio, which can be used to normalize the images and make them easier to compare.

4. CNN model - Classifier

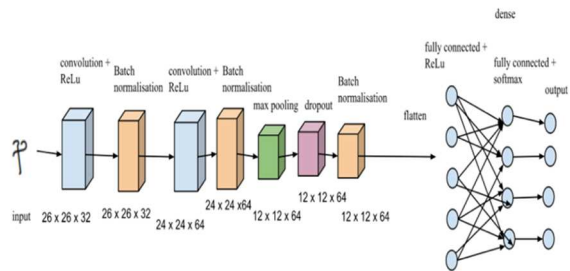


Fig 4.6: CNN architecture

A Convolutional Neural Network predicts the character from segmented character images[15]. Input to the CNN model is a matrix of pixels of size 28x28.

This CNN architecture uses two convolutional 2D layers, batch normalization layers, a max-pooling layer, and ReLu and Softmax functions. The first convolutional 2D layer has 32 output filters with a kernel size of 3x3 (height and width of sliding window). The ReLu activation function is applied to the layer. The output size of this layer is 26x26. The input tensor and the element-wise maximum of 0 are returned by the ReLu (Rectified Linear Unit) activation function. Therefore, the ReLu function helps to prevent exponential growth in the computation of convolutional layers. ReLUs also prevent the "vanishing gradient" problem.

A batch normalization layer is used after the convolutional layer. The mean and standard deviation of the most recent batch of inputs are used by the layer to normalize its output during training. The layer uses a moving average of the mean and standard deviation of the clusters seen during training to normalize its output during inference.

The second convolutional 2D layer of 64 output filters and ReLu activation function is used. The output size of this layer is 24x24. A batch normalization layer is used after this layer. A max-pooling 2D layer has a pool size of (2,2), which will take the max value over a 2x2 pooling window to downsample the input. The output size of the max pooling layer is 12x12.

The dropout layer is used with a rate of 0.25. At each training phase, the dropout layer randomly sets input units to 0 with a frequency of 0.25 in order to prevent overfitting.

The Dropout layer is a mask that leaves all other neurons unaltered while eliminating particular neurons' contribution to the following layer. It is applied to hidden layers.

A batch normalization layer is applied to the output of the dropout layer. A flattening layer is used in which the flatten function reduces the incoming tensors from multiple dimensions to just one dimension so that a neural network layer of $(12 \times 12 \times 64 = 9216)$ neurons are created to handle each element of incoming data.

Each neuron in the dense layer gets input from all the neurons in the layer below it because it is a completely connected layer. A unit, or the number of nodes in the layer, and activation are the two parameters required by the dense function. The element-wise activation function is applied in a thick layer with the aid of the activation parameter.

In our CNN architecture, the first dense layer has 256 nodes in the layer, and the ReLu activation function is applied. The second dense layer has 216 nodes, and a softmax activation function is applied. Unit 216 is chosen because there are 216 classes in the training dataset.

A vector of numbers is transformed into a probability distribution by Softmax. A classification network's final layer is activated by Softmax because the outcome can be read as a probability distribution.

Loss

Category cross-entropy is the chosen loss function. A classification model that classifies the data by estimating the likelihood that it corresponds to one class or the other is trained using an optimization function called the cross entropy loss function.

5. Digitizing the script using Vatteluttu TrueType Font Transliterated text is obtained in the previous step. To digitize the Vatteluttu script, we used a Vatteluttu TTF. Using this TrueType Font, the system can recognize and output the Vatteluttu characters onto the standard output.

V. RESULTS AND DISCUSSION

The stone and manuscript images are given as input to obtain editable text in Vatteluttu, Tamil, and Devanagari scripts. The result obtained from the CNN model has produced an 82.14% recognition rate after training with augmented data.

With the initial dataset we had, the training of the CNN model resulted in an accuracy of 6.96 percent, which was too low. The reason for this low accuracy was not having enough datasets. The dataset's size was increased by data augmentation, which further increased accuracy to 82.14 percent.

For the convolutional neural networks to be trained successfully, hyperparameter adjustment is necessary. The hyperparameters are determined by the neural network's architecture and can be adjusted using various variables for better outcomes.

The batch size is 128, which was determined to be ideal for the suggested model. With maximum accuracy, the suggested model can run up to 70 epochs.

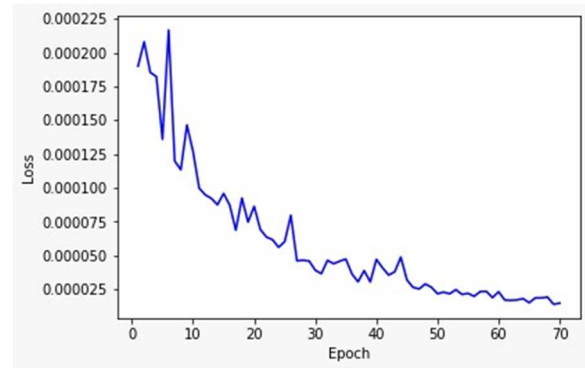


Fig 5.1: Loss of CNN model plotted in Epoch vs Loss graph

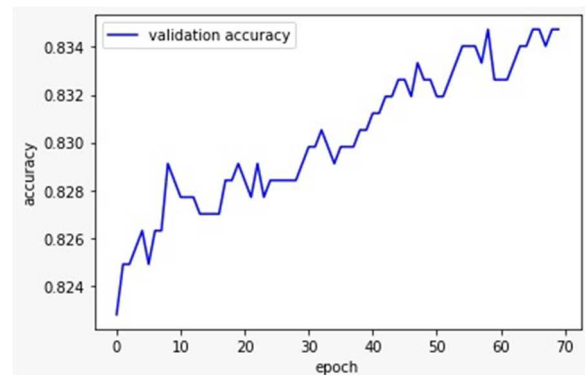


Fig 5.3: Accuracy of CNN model plotted in epoch vs. accuracy graph

VI. CONCLUSION

The proposed work uses Neural networks to extract information to recognize ancient Tamil scripts. The outcomes of the prehistoric character identification task using the CNN model demonstrate that accurate categorization is feasible when employing deep neural networks. By using a larger dataset and an enhanced neural network method for recognition, this system can overcome the problems plaguing the earlier designs. The outcomes also highlight the need for improvement, particularly in Tamil character recognition.

[illegible]

Fig 5.2: Table of an input image, the predicted Vatteluttu text, and its transliterated version in Tamil and Devanagari script

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